Problem Statement:

E-commerce company wanted a better understanding of their customer's shopping behavior by analyzing review rating, frequency of repeaters and overall average customers visiting their website to improve their sales. Sales team has observed a drift in purchasing patterns across product verticals. They are specifically looking into factors such as discounts, reviews, purchase season or payment preferences to drive consumer decisions and repeat purchases.

You as a data analyst tasked with analyzing the company's customer behavior dataset to answer the following business problem statement:

"How can the company use customer shopping data to identify trends to improve customer engagement, and optimizing marketing product strategies?"

Deliverables:

I. **Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
II. **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
III. **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
IV. **Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
V. **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.